

FromDonna Runtime Persistence

Technical Report — Architecture B (Worker Pull → R2)
Investigation · Design · Implementation · Live Verification · Updated
Repo: FromDonna · July 2026 · Status: Live & verified on production traffic

Status (as of live tests)

Architecture B is live and working: after agent use, the sandbox stages a filtered checkpoint; the Worker pulls it into R2 (harness export and/or E2B envd). On missing/replaced E2B, the Worker restores from R2. Verified for user telegram:1063008785 with multiple auto updates (source envd-pull / gateway-session). Channel-agnostic keys. Secrets excluded.

1. Executive summary

FromDonna runs one E2B sandbox per product user. Hermes state lives under `/home/user/.hermes` (agent home) and working files under `/home/user/workspace`. Normal E2B **pause/resume does not clear** that disk. The product risk is hard loss: when the sandbox is missing or **replaceRuntime** creates a new VM, prior state would vanish without a durable copy.

This project adds **Architecture B**: stage checkpoint on the sandbox after agent session completion, Worker **pulls** into Cloudflare R2, and restores on create/replace. Goal is proactive backup after agent use so state can be restored if E2B is gone—not to replace pause/resume for idle.

Delivered and live:

- R2 bucket `fromdonna-user-state`
- Worker `fromdonna-telegram-gateway` with `USER_STATE` binding + harvest/pull/restore
- Template `fromdonna-hermes` with stage + export endpoints
- Live auto-backups observed (`envd-pull` / `gateway-session`) for test user

2. Problem & product decisions

2.1 When restore is needed

Event	Same disk?	Needs R2?
Idle pause → unpause	Yes	No
connect 404 / broken resume → replaceRuntime	No	Yes
Failed / stuck provision → new create	No	Yes if prior state
Planned kill / template rebuild	No	Yes

2.2 Locked decisions

Decision	Choice
Survive hard loss	Agent home + sessions + workspace
Save policy	Proactive after agent use; next-message / pre-replace pull if post-turn races
Architecture	B — stage on sandbox, Worker pulls to R2
Secrets on R2	Never (<code>.env</code> , <code>auth.json</code> , keys excluded)
Channel naming	Agnostic <code>users/{userId}/...</code>
Idle path	E2B <code>autoPause/resume</code> unchanged

3. Why not sandbox → Worker push (original simple plan)

Original plan: after turn, sandbox POSTs tar to Worker `/internal/checkpoint`, Worker writes R2. In production, `E2B→*.workers.dev` POSTs returned **Cloudflare error 1010** (bot/signature block). That is an edge policy issue, not an R2

issue.

Architecture B keeps R2 credentials only on the Worker, avoids opening broad bot exemptions, and uses Worker→sandbox pulls (already trusted for inject/bootstrap).

Option	Pros	Cons
A Push sandbox→Worker	Simple post-turn semantics	1010 without CF bot/WAF/custom-domain changes
B Pull (chosen)	Works with current CF; multi-path harvest	More parts; async timing
C Presigned PUT→R2	Post-turn + no long-lived keys in box	Needs R2 S3 API tokens + presign code

4. Architecture B (implemented)

4.1 Flows

Stage (sandbox, after agent session):

Gateway session finishes → pack filtered agent-home + workspace →
/home/user/.hermes/fromdonna-checkpoint-latest.tar.gz + ready marker

Harvest (Worker):

- 1) Separate waitUntil after inject (does not starve create/inject budget)
- 2) Poll harness GET /internal/checkpoint/export
- 3) Fallback: E2B envd GET /files for staged tar
- 4) Also pull at start of next message and before replace/kill

Restore (Worker on create/replace):

R2 → POST harness /internal/restore → extract agent-home + workspace

4.2 R2 layout (channel-agnostic)

users/{userId}/checkpoint.tar.gz
users/{userId}/manifests/latest.json

Manifest: version, userId, savedAt, bytes, sha256, runtimeId, source

4.3 Pack filter (exclude)

- Secrets: .env, auth.json, key/token-like files
- Regenerable: caches, venvs, node_modules, checkpoints, backups, WAL/SHM
- Runtime noise: pids, turn locks, gateway_state.json, staged checkpoint files themselves
- SQLite: consistent *.db via sqlite3.backup()
- Never: /opt/fromdonna install tree

4.4 Source labels (observed)

source	Meaning
gateway-session	Staged after Hermes gateway session; often via harness export path
envd-pull	Worker read staged tar via E2B envd filesystem API
harness-export	Worker GET /internal/checkpoint/export
manual-worker-pull	Ops/manual completion during bring-up

5. Implementation map

Area	Files / pieces
Worker	src/checkpoint.ts, src/index.ts (harvest, pull, restore), wrangler.toml USER_STATE
Harness	checkpoint.py (pack/stage/export/restore), server.py, gateway_runtime.py
Template	fromdonna-hermes (warm uvicorn + harness)
Docs	memorymanagement.md; this PDF

Endpoint	Role
POST /internal/checkpoint	Optional push (blocked by 1010 from E2B; kept for ops/tests)
GET /internal/checkpoint/status	Ops: exists/bytes/manifest
GET harness /internal/checkpoint/export	Worker pull staged tar
POST harness /internal/restore	Worker pushes R2 blob into new runtime

6. Production deployment

Component	Status
R2 bucket fromdonna-user-state	Created & bound as USER_STATE
Worker fromdonna-telegram-gateway	Deployed with harvest + envd fallback
Template fromdonna-hermes	Rebuilt with stage/export/restore
Public health	OK

Live Worker: <https://fromdonna-telegram-gateway.code-df4.workers.dev>

7. Live verification (telegram:1063008785)

Multiple automatic R2 writes observed after real Telegram turns on runtime iujkhcndfuui0u8td7c1m.

When (UTC)	bytes	source	Notes
~03:39	~8.8 MB	manual-worker-pull	Bring-up: stage worked; first auto race missed
04:00:02	9,070,390	envd-pull	First clear auto path success
04:11:36	9,173,391	gateway-session	Test after reply — PASS (updated sha256)
05:14:21	9,312,881	envd-pull	Later recheck — UPDATED again

Test protocol used

- 1) Record baseline manifest savedAt/sha256
- 2) User texts agent and receives reply
- 3) Wait ~1–2 minutes for stage + harvest
- 4) Compare R2 manifest — PASS if newer savedAt and/or changed sha256

On-box signals: staged tar under `~/hermes/fromdonna-checkpoint-latest.tar.gz`; ready marker may clear after successful export consume.

8. Reliability model (honest)

Expectation	Status
Stage after agent session (new image)	Working
Automatic land in R2 after use	Working (envd-pull / export); multi-path
Instant same-second as Telegram reply	Not guaranteed (async harvest)
Next-message / pre-replace pull safety net	Implemented
Restore if E2B missing	Implemented on provision/replaceRuntime
Pause/unpause needs R2	No — disk retained

Primary success criterion for the product: **after regular agent use, a durable R2 copy exists so a future missing sandbox can be restored**. Live traffic on the test user meets that criterion.

9. Security

Asset	Handling
Telegram bot token	Worker only (Bot API proxy)
E2B API key	Worker only (connect/files for pull)
Harness secret	Worker + sandbox control plane
R2 credentials	Worker binding only (no long-lived R2 keys in sandbox)
Checkpoint contents	Filtered; no .env / auth.json
Size cap	40 MiB per checkpoint body

10. Failure modes & mitigations

Failure	Mitigation
CF 1010 on sandbox→Worker POST	Do not rely on push; use Worker pull (B)

Failure	Mitigation
Harvest races long create/inject	Separate waitUntil; next-message pull; pre-replace pull
Export 204 / not ready	Poll + envd fallback; tar-only export if marker missing
Restore fail	Soft-fail log; runtime still usable empty
Brand-new user, no prior R2	Empty start is correct

11. Follow-ons (optional)

- Incremental sync / last-K versions
- Presigned R2 PUT (Arch C) if pure post-turn push is desired without bot fight
- Metrics: harvest success rate, restore hits, replaceRuntime count
- Force-migrate remaining old sandboxes to latest template
- Product media tools fully on R2 (r2:// schema already exists)

12. Code touch list

```
cloudflare/gateway/
  wrangler.toml — R2 USER_STATE
  src/checkpoint.ts — put/get/manifest/auth
  src/index.ts — restore; harvestCheckpointAfterTurn; harness+envd pull; pre-replace pull
E2B-Template/harness/
  checkpoint.py — pack/stage/export/restore filters
  server.py — /internal/restore, /internal/checkpoint/export
  gateway_runtime.py — stage after session complete
documentation/deployment/
  memorymanagement.md
  fromdonna-persistence-technical-report.pdf — this document
```

13. Bottom line

Question	Answer
Does unpause wipe memory/workspace?	No.
What architecture?	B: stage on box → Worker pull → R2 → restore on missing/replace.
Is it live?	Yes.
Does post-use backup work?	Yes — verified multiple times on real Telegram turns.
Why not simple push?	E2B→workers.dev blocked by CF 1010; pull is reliable without loosening bots.

Updated after production verification of Architecture B (Worker pull via harness export and envd). Grounded in FromDonna gateway, harness, E2B template, and live R2 objects for telegram:1063008785.